



TAMPINES JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2008

MATHEMATICS

9740/02

Higher 2

Paper 2

Tuesday

26 August 2008

3 hours

Additional materials: Answer paper
 Cover Page
 List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in, including the Cover Page.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Do not write anything on the List of Formulae (MF15).

Answer **all** the questions.

Begin each answer on a **fresh page** of paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are required of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, place the completed cover page on top of your answer scripts and fasten all your work securely together with the string provided.

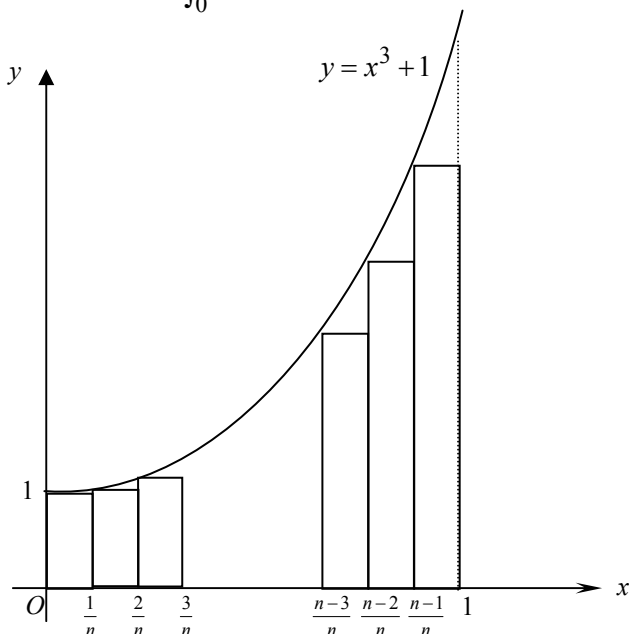
For Candidate's Use	For Examiner's Use
Question Number	Marks Obtained
1	
2	
3	
4	
5	
6	
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9	
10	
11	
12	
Total	
Calculator Model:	

This question paper consists of 7 printed pages including this page.

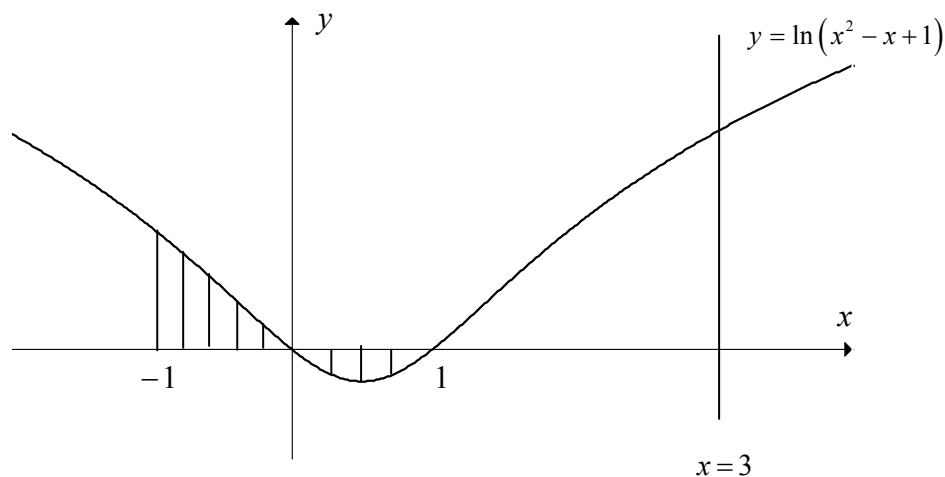
Section A: Pure Mathematics [40 marks]

- 1** Function f is defined by $f: x \mapsto 1 + \frac{1}{x}$ for $x > 0$. State the range of f and find the inverse function f^{-1} .
The composite function g is defined as $f \circ f$. Define g in a similar form and determine whether there exist any values of x such that $g(x) = 0$. [6]
- 2** (a) The first term of an arithmetic series is 7, its last term is 70 and its sum is 385. Find the number of terms in the series and the common difference. [3]
- (b) A geometric series is given as follows:
- $$2 - (x+1) + \frac{(x+1)^2}{2} - \frac{(x+1)^3}{2^2} + \dots + \frac{[-(x+1)]^{r-1}}{2^{r-2}} + \dots$$
- Find the set of values of x such that the sum to infinity of the series exists and the value of x for which the sum to infinity of the series is $\frac{4}{3}$. [4]
- 3** The points A , B and C are given by $\overrightarrow{OA} = 2\mathbf{i} + 4\mathbf{j} - 4\mathbf{k}$, $\overrightarrow{OB} = 4\mathbf{i} + 5\mathbf{j}$ and $\overrightarrow{OC} = -\mathbf{i} - 2\mathbf{j} - \mathbf{k}$. The point D on AC is such that $2\overrightarrow{AD} = \overrightarrow{DC}$ and l is the line through D perpendicular to AB and AC .
- (i) Show that the vector equation of l is $\mathbf{r} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k} + \alpha(-3\mathbf{i} + 2\mathbf{j} + \mathbf{k})$ where α is a parameter. [3]
- (ii) Given that P is a point on l such that $PD = \sqrt{14}$. Find the possible position vectors of P . [4]
- 4** A sequence is defined by $u_{n+1} = u_n + 2^n$, $u_1 = 3$.
- (i) Find u_n for $n = 2, 3, 4, 5$. [2]
- (ii) By considering $u_n - 1$, give a conjecture for u_n in terms of n . [1]
- (iii) Prove your conjecture by induction for $n \geq 1$. [4]

- 5 (a) The diagram shows part of the graph of $y = x^3 + 1$ from $x = 0$ to $x = 1$. The area under the curve may be approximated by the total area, A , of n rectangles as shown. The width of each rectangle is $\frac{1}{n}$. Given that $\sum_{r=1}^n r^3 = \left[\frac{n(n+1)}{2} \right]^2$, show that $A = 1 + \frac{1}{4} \left(1 - \frac{1}{n} \right)^2$. Hence deduce the exact value of $\int_0^1 x^3 + 1 \, dx$. [5]



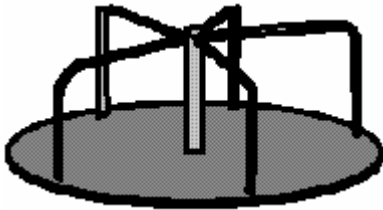
- (b) The diagram shows the curve with equation $y = \ln(x^2 - x + 1)$.
- (i) Find the total area of the shaded regions. [3]
- (ii) Find the volume of the solid formed when the region bounded by the curve, x -axis and the line $x = 3$ is rotated through 2π radians about the y -axis. [5]



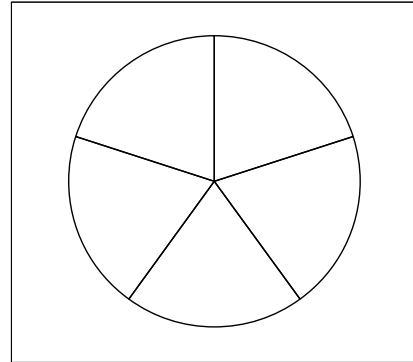
Section B: Statistics [60 marks]

- 6 A country club has 1000 members of which 800 are males and 200 are females. Their names are recorded in a computer system in alphabetical order. The committee of the club wishes to find out the type of activities preferred by the members and conducts a survey on 100 members of the club.
- (i) Explain how a systematic sampling method can be used to select these 100 members. [2]
 - (ii) Give one advantage and one disadvantage of this method. [2]
 - (iii) Suggest a method of obtaining a more representative sample and describe how it may be carried out. [2]

- 7 A rotating platform with five identical sectors is installed in a carnival. Each sector can take up to two persons.



The figure shows a rotating platform



The figure shows the top view of the platform

A group of 4 friends and 3 individuals boarded the platform and occupied all the sectors.
[For the following questions, you do not need to consider the arrangement within each sector]

- (i) If the 4 friends requested that they will stand in 2 adjacent sectors, find the number of ways in which this group of people can be arranged. [2]
- (ii) If the 4 friends must stand in 4 different sectors, find the number of ways in which this group of people can be arranged. [4]

- 8 (a) Comment on the validity of the following statement about the product moment correlation coefficient, r , of a bivariate sample (x_i, y_i) , $i = 1, \dots, n$.

If X is the number of late-coming of a student in a year and Y is the student's performance in the Mathematics Examination, then $r \approx -1$ implies that late-coming causes poor performance in the subject. [1]

- (b) A group of researchers would like to study the relationship between the width of forefinger of a person's master hand and his speed of typing. Ten people randomly selected from all walks of life were required to type a letter which contains more than 200 words. They were asked to type as many words within a two minute interval. The width of forefinger on the master hand of each person was measured and recorded to the nearest 0.1 cm. The following are the results:

Width (cm) of forefinger, x	1.0	1.1	1.1	1.3	1.4	1.5	1.6	1.8	1.8	1.9
No. of Words (per 2 mins), y	30	36	46	58	66	78	90	106	120	134

- (i) It is given that the value of the product moment correlation coefficient for the above data is 0.987, correct to 3 decimal places. State what its value tells you about the likely relationship between width of forefinger of a person's master hand and his speed of typing. [1]
- (ii) Obtain both the equations of least squares regression lines of y on x and x on y . [2]
- (iii) Estimate x when $y = 100$, giving your answer to 2 decimal place. Comment on the reliability of your result. [2]
- (iv) The width of forefinger and the number of words that a person can type are re-recorded in millimetres and number of words per minute. State, giving a reason, whether the value of r in (i) would change. [1]

- 9 A research study indicates that 16% of the total population are more than 70 years old and 18% of the total population have a measurable hearing defect. Of those more than 70 years old, 65% have a measurable hearing defect. Find, correct to 3 decimal places, the probability that a person chosen at random from the population is

- (i) more than 70 years old and has a measurable hearing defect, [1]
- (ii) less than 70 years old and has a measurable hearing defect, [2]
- (iii) either more than 70 years old or has a measurable hearing defect or both, [2]
- (iv) more than 70 years old given that the person has no measurable hearing defect. [3]

- 10** The weights of girls and boys in a college have independent normal distributions with their means and standard deviations given in the following table.

Gender	Mean	Standard deviation
Girls	44 kg	5 kg
Boys	50 kg	4 kg

- (a) Find the greatest weight (correct to nearest 0.1 kg) exceeded by at least 10% of the boys. [2]
- (b) Find the probability that the difference of the weights of 2 randomly chosen girls is less than 1.2 kg. [2]
- (c) One girl and one boy are chosen at random. Find the probability that one of them is at least 47 kg and the other is less than 47 kg. [3]
- (d) Find the probability that the average weight of 2 randomly chosen girls and 8 randomly chosen boys is less than 50 kg. [3]

- 11** On average, 1 in 20 domestic flights from a particular airport is delayed.

- (i) Find the probability that there are less than 2 domestic flight delays out of 10 randomly chosen domestic flights. [2]
- (ii) Using a Poisson approximation, find the probability that there are at least 56 domestic flights arriving on time out of 60 randomly chosen domestic flights. [3]

It is also known that the number of international flight delays from the same airport is a Poisson variable with an average rate of 1 in 2 days. Using suitable approximations, find the probability that

- (iii) the number of international flight delays in 30 days exceeds 16. [3]
- (iv) the mean number of international flight delays per day, in a period of 60 days, is less than 0.7. [3]

- 12** A beauty chain wants to advertise about its one-month slimming programme. It claims that the mean weight loss of a woman after going through its one-month slimming programme is at least w kg. A random sample of 60 women was selected and their weights before and right after the completion of the programme were recorded. The weight loss, X , in kg of each woman was computed and summarized by $\sum x = 405$ and $\sum x^2 = 8495.529$.

- (i) Find unbiased estimates of the population mean and variance. [2]
- (ii) Show that the greatest value of w is 8 (correct to the nearest kg) if the result of the test, performed at 5% level of significance, supports the beauty chain's claim. [5]

A particular outlet of the beauty chain claims to be meeting the expectation of its chain. The weights of 10 randomly chosen women taken before and right after the completion of the programme and the weight loss, X in kg of each woman was computed and given below:

7.7 7.7 7.8 7.8 7.9 8.0 8.0 8.0 8.1 8.2

Taking w to be 8 and test whether the outlet's claim is valid at 5% level of significance. You should state any assumptions that you need to make. [5]